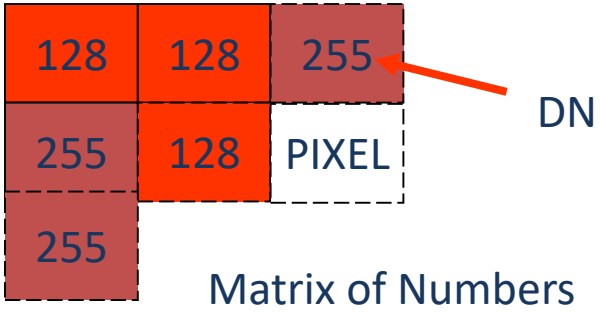

بسم الله الرحمن الرحیم

DIGITAL IMAGE AND RESOLUTIONS

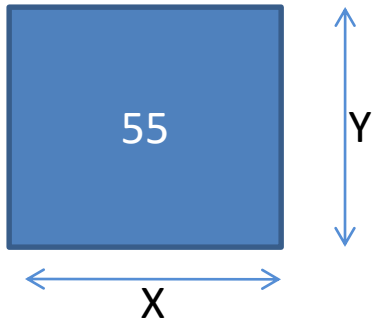
DIGITAL IMAGE

- **Digital images consist of discrete elements called PIXELS**
- **Size of this area affects the details within the scene**
- **As the pixel size is reduced, more scene detail is preserved**



Pixel

- **A pixel has both spatial and spectral properties**
  - **The spatial property defines the dimensions of the corresponding ground area**
  - **The spectral property defines the intensity of the spectral response for a pixel in a particular band**
  - **Aspect Ratio :1**



RESOLUTION

- Is the ability of an entire remote-sensing system to render a sharply defined image
- It indicates the smallest observable (measurable) difference
- Image resolution describes the details in an image
- Types
  - Spatial, Spectral, and Radiometric- properties of RS instruments
  - Temporal: property of platform

SPATIAL RESOLUTION

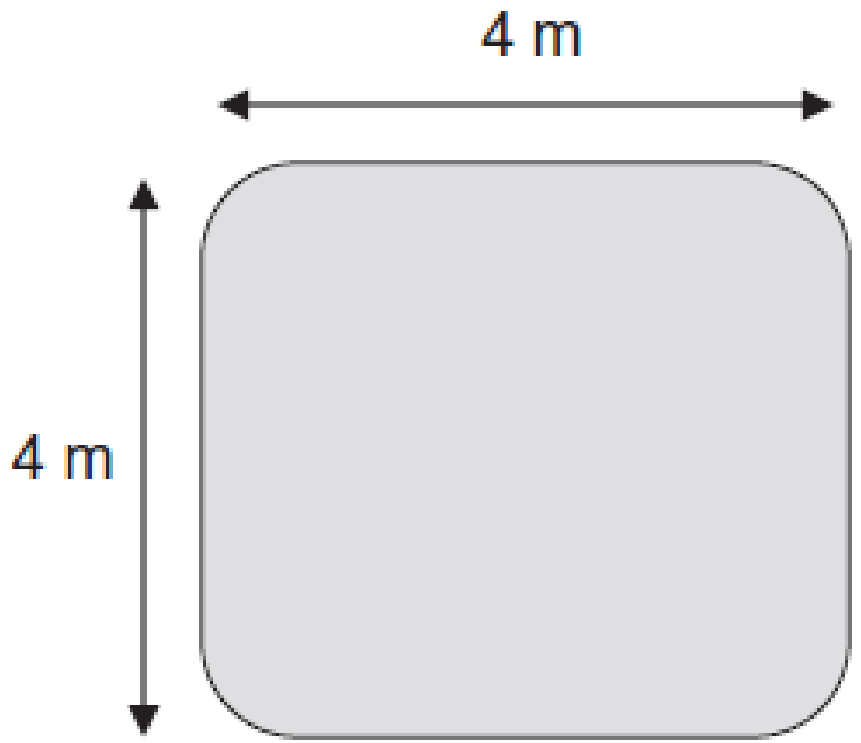
- **Spatial resolution measures the smallest object that a satellite sensor can detect.**
- **It is an indication of how well a sensor can record spatial details**
- **Also refers to the size of the area on the ground represented by each pixel (area covered by a pixel)**
- **In raster images, the smaller the area of land that each pixel represents, the higher the resolution of the data, the more the ground details, and the larger the files needed to store the data**

SPATIAL RESOLUTION

- **To be detected, an object must be equal to or larger than the resolution and spectrally different from its surrounding objects.**
- **Smaller may be detectable if their reflectance is dominant**
- **To detect an object, generally, the spatial resolution should be <1/2 size of a smaller dimension**

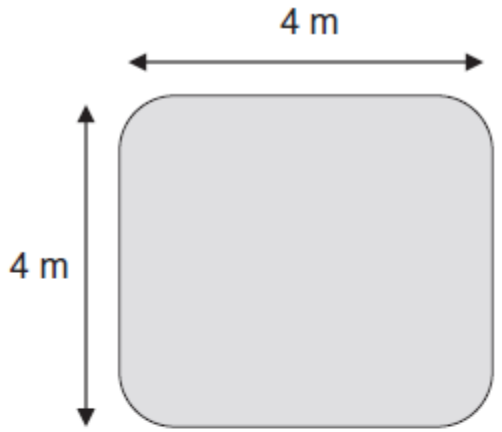
SPATIAL RESOLUTION

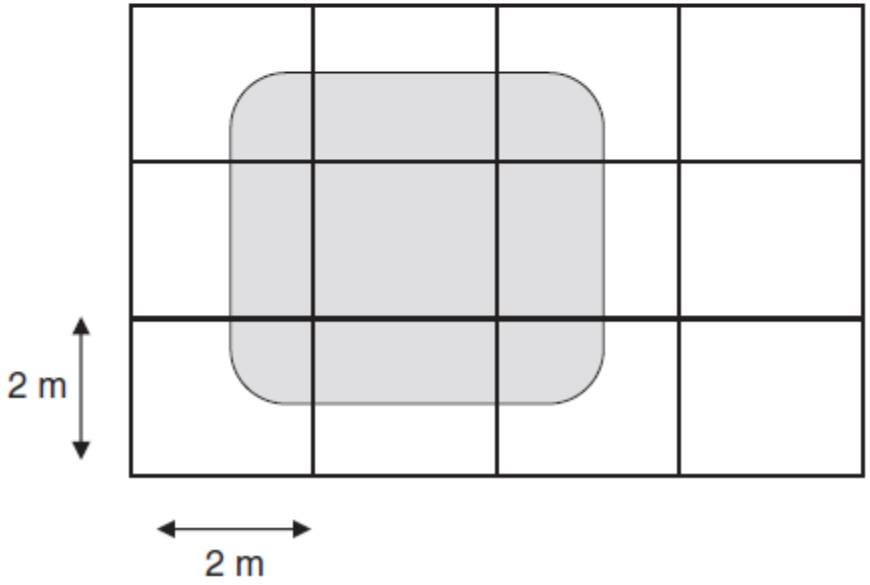
- **If we need to identify a home that is 4 m wide, the minimum spatial resolution of the imagery would be 2 m**



SPATIAL RESOLUTION

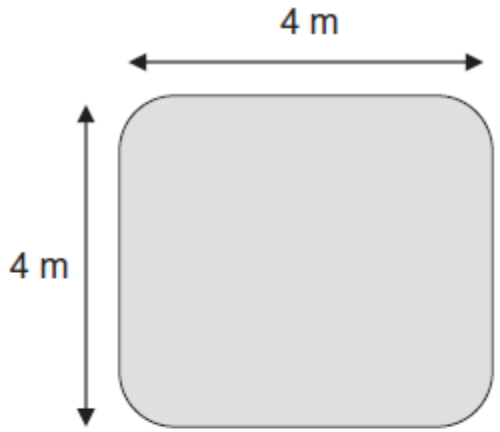
- **In a real world situation, a 4 m wide object to be identified in an image most likely does not locate perfectly over 4 pixels in a 2 m resolution image**



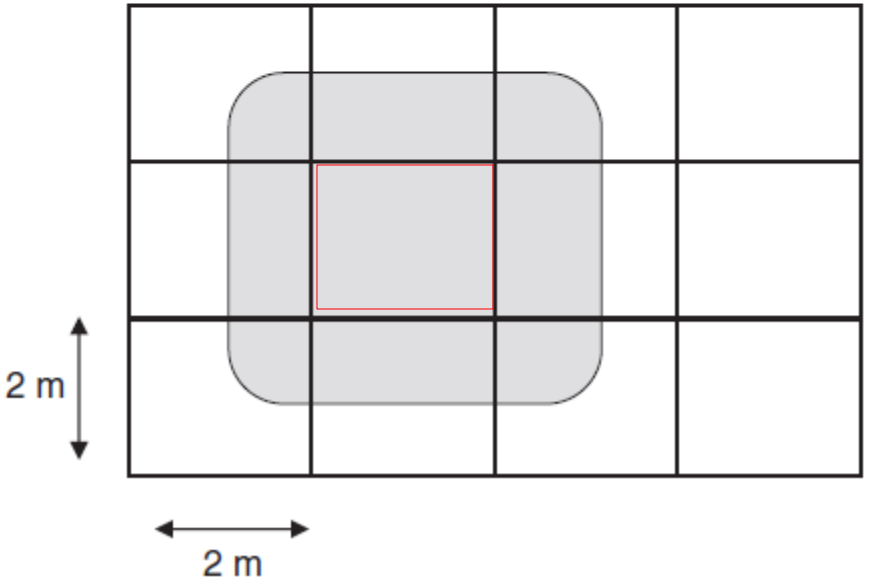


SPATIAL RESOLUTION

A 4-m wide object and 2m resolution image data

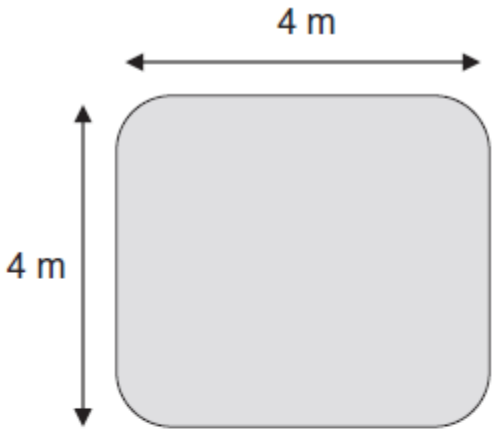


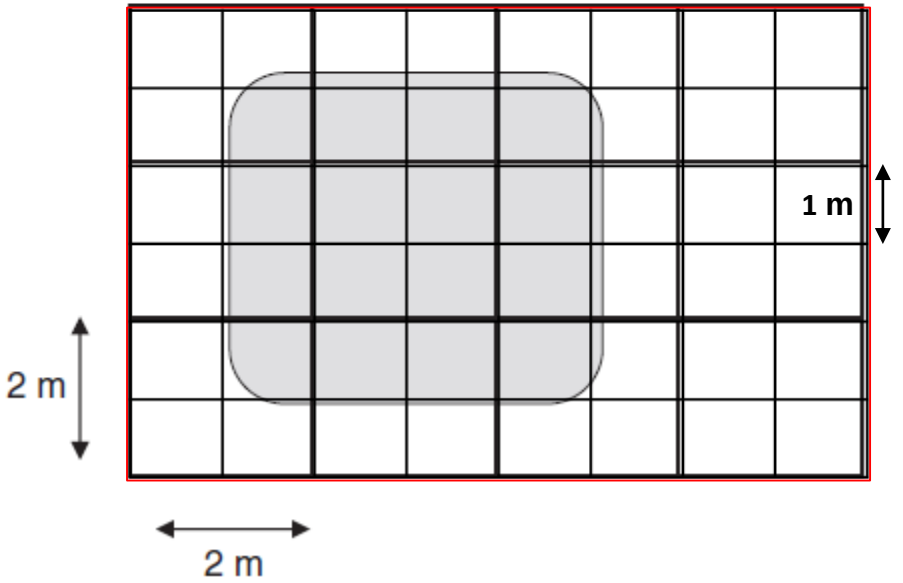
There is only 1 pixel that contains a pure spectral response from the object



SPATIAL RESOLUTION

- Hence, we need a pixel size that is remarkably smaller than an object to identify that object in an image
- Since urban objects are notably smaller in comparison to natural features, it is apparent that we need a significantly small pixel size for urban applications





SPATIAL RESOLUTION

- **Coarse/low Resolution- only large features are visible**
  - **> 30 meters to > 1000 meters**
- **Medium Resolution**
  - **> 4 meters to 30 meters**
- **High /fine Resolution- small objects can be detected, and more ground details**
  - **1 to 4 meters**
- **Very High Resolution –sub-meter**

HIGH RESOLUTION SATELLITE IMAGES



WorldView-2



GeoEye-1

HIGH RESOLUTION AERIAL IMAGES

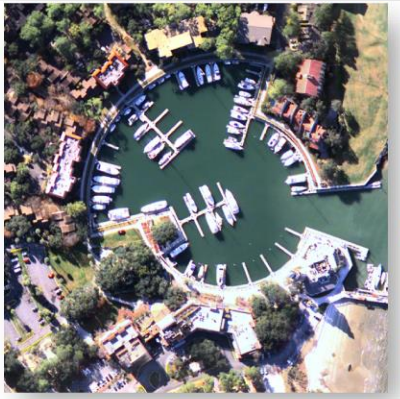


ADS-80, 10-25 cm

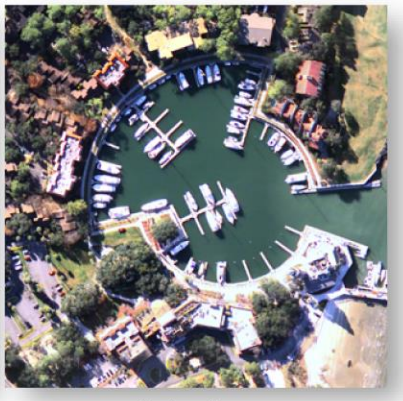


ULTRACAM$_D$, 12 cm

Spatial Resolution



a. 0.5 x 0.5 m.



b. 1 x 1 m.



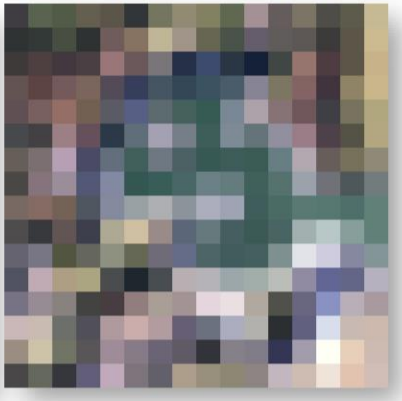
c. 2.5 x 2.5 m.



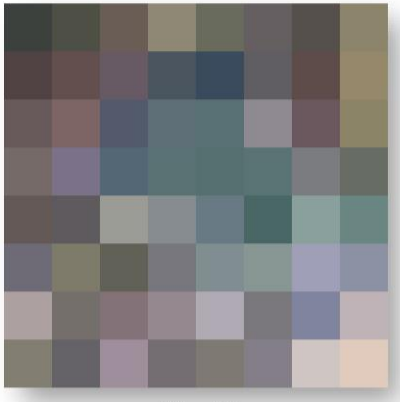
d. 5 x 5 m.



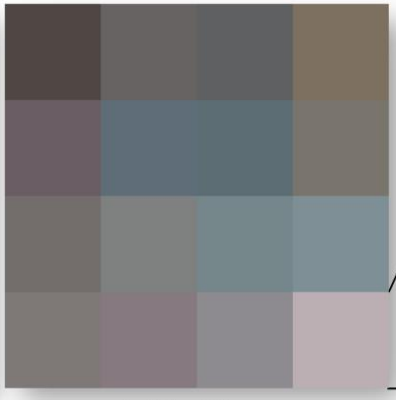
e. 10 x 10 m.



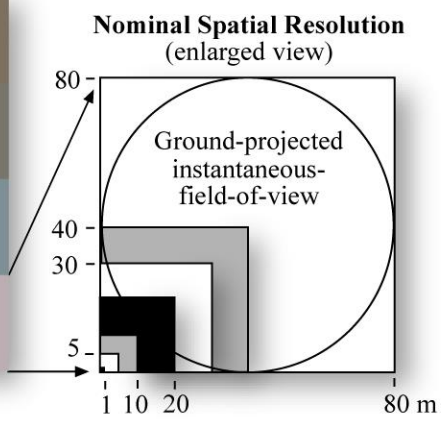
f. 20 x 20 m.



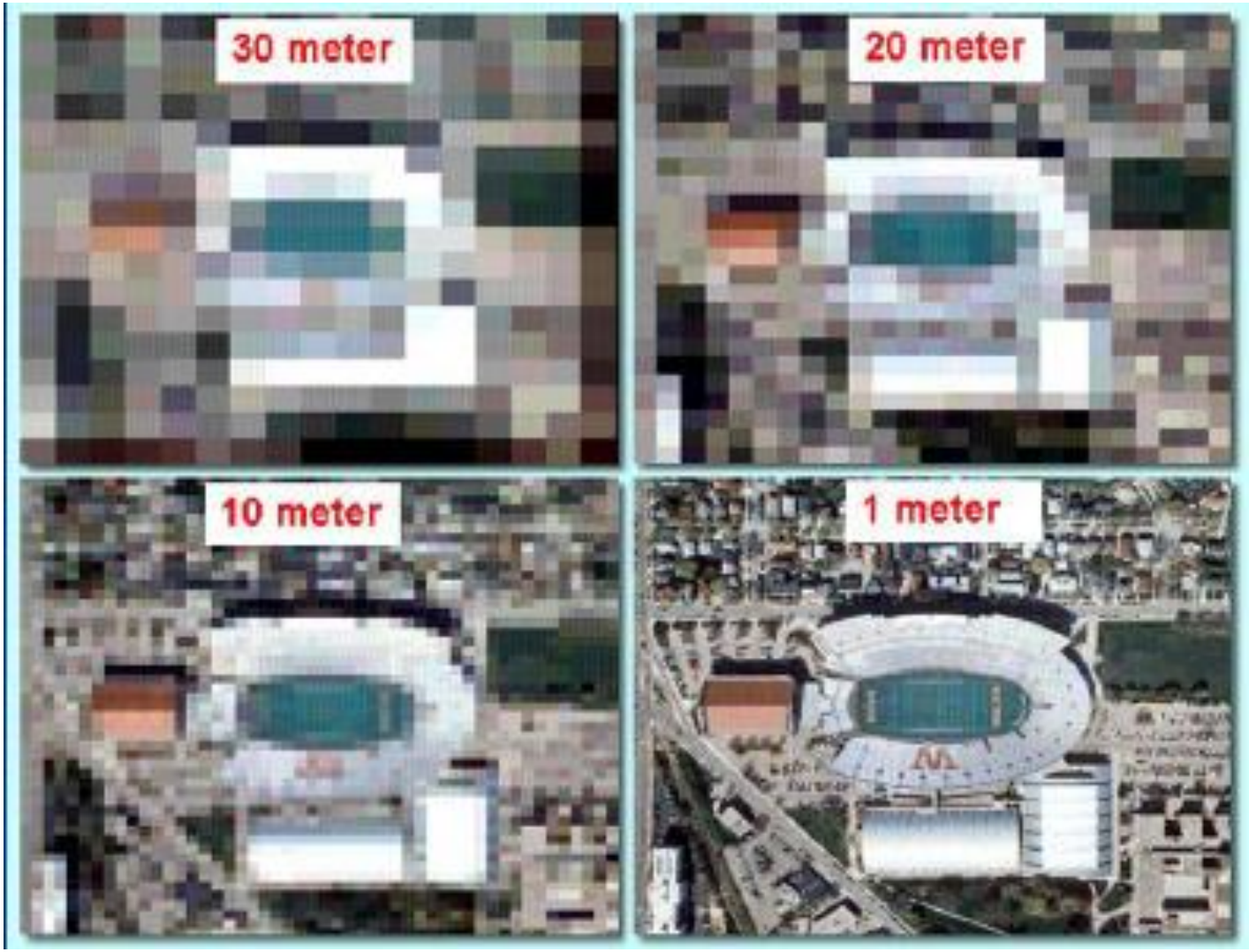
g. 40 x 40 m.



h. 80 x 80 m.



SPATIAL RESOLUTION



Map Scale

- **The ratio of distance on an image or map to actual ground distance**
- **Maps or images with small "map-to-ground ratios" are referred to as small scale (e.g., 1:100,000),**
- **Those with larger ratios (e.g., 1:5,000) are called large scale maps**

Map Scale Vs Details

Scale 1:500,000



1 cm = 5 kilometer

Scale 1:200,000



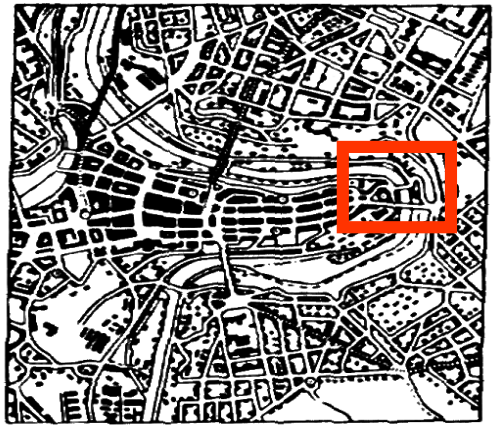
1 cm = 2 kilometer

Scale 1:100,000



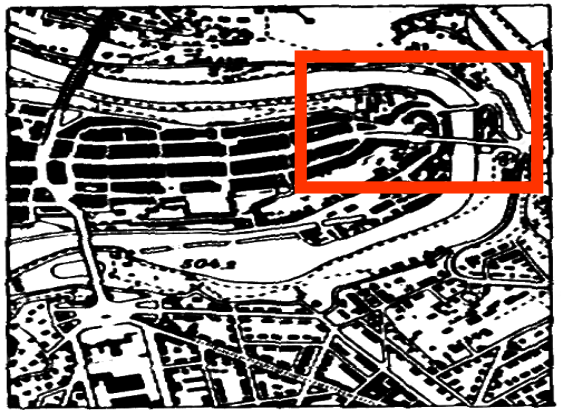
1 cm = 1 kilometer

Scale 1:50,000



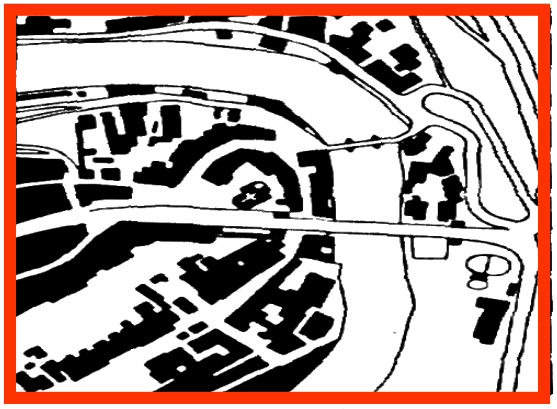
1 cm = 500 meters

Scale 1:25,000



1 cm = 250 meters

Scale 1:10,000



1 cm = 100 meters

Scale Vs Spatial Resolution (Pixel Size)

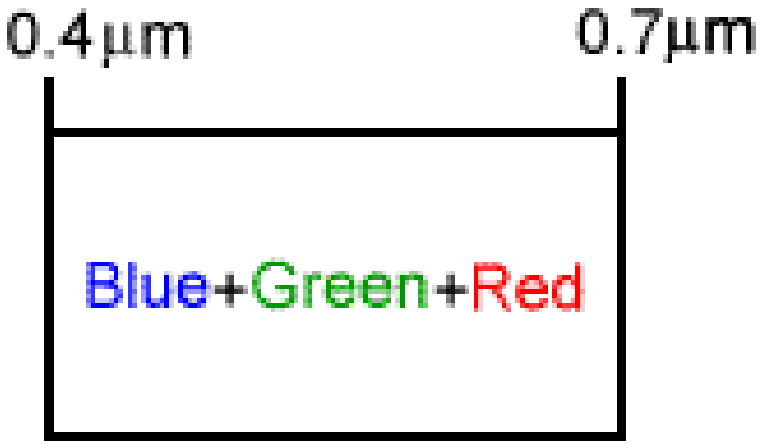
Scale	Approx Pixel Size (M)	Sensor
1: 50,000	4	Aircraft, IKONOS
1: 250,000	20-30	SPOT, Landsat TM
1: 500,000	50	Landsat MSS
1: 10,000,000	1000	NOAA AVHRR

SPECTRAL RESOLUTION

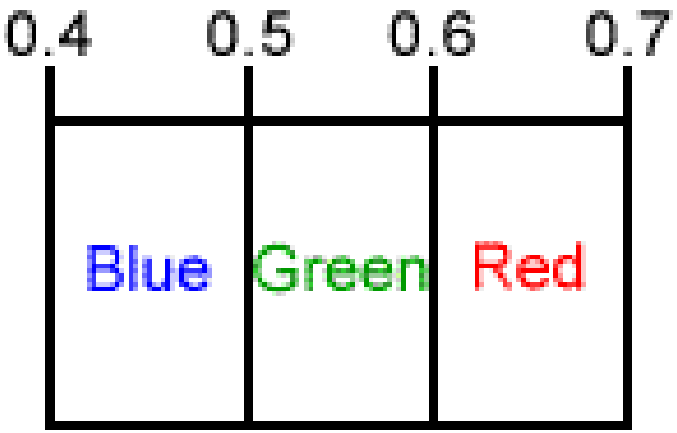
- It refers to the specific wavelength intervals in the electromagnetic spectrum that a sensor can record or is sensitive to
- Each wavelength interval is referred as band, which records a specific portion of the electromagnetic spectrum
- High spectral resolution is achieved by narrow band widths
- The finer the spectral resolution, the narrower the wavelength range for a particular band
- In general, it is the no of bands of the remote sensing sensor system

SPECTRAL RESOLUTION

- **Is the ability of a sensor to define fine wavelength intervals**



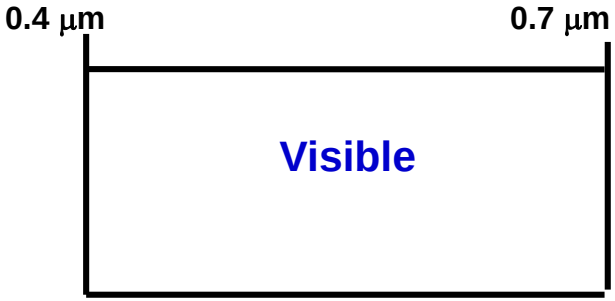
Panchromatic
One Band
Gray Scale



Multispectral
Three Bands
Color Image

Spectral Resolution

- Example: gray scale image
  - Single band
  - Intensity is sum of intensity of all visible wavelengths





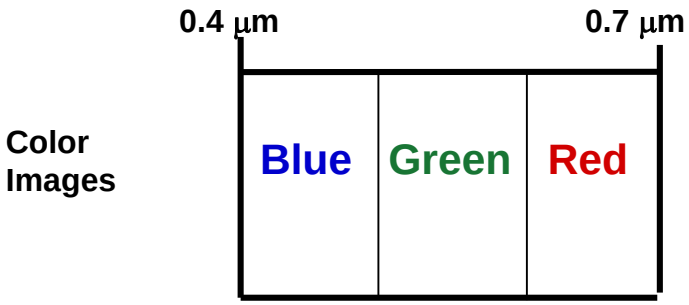
Can you tell the color of the platform top?

How about color of sash?

Spectral Resolution

- Example: Color image

Color images need least three bands, e.g., red, green, and blue; RGB



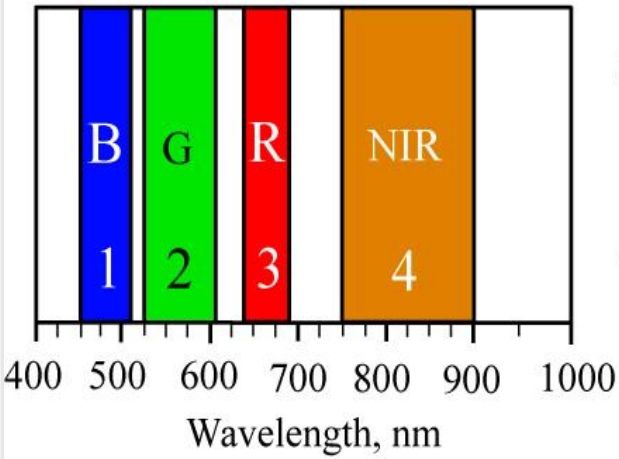
- Using increased spectral resolution (three sensing wavelengths) adds information

- In this case by “sensing” RGB can combine to get full color rendition

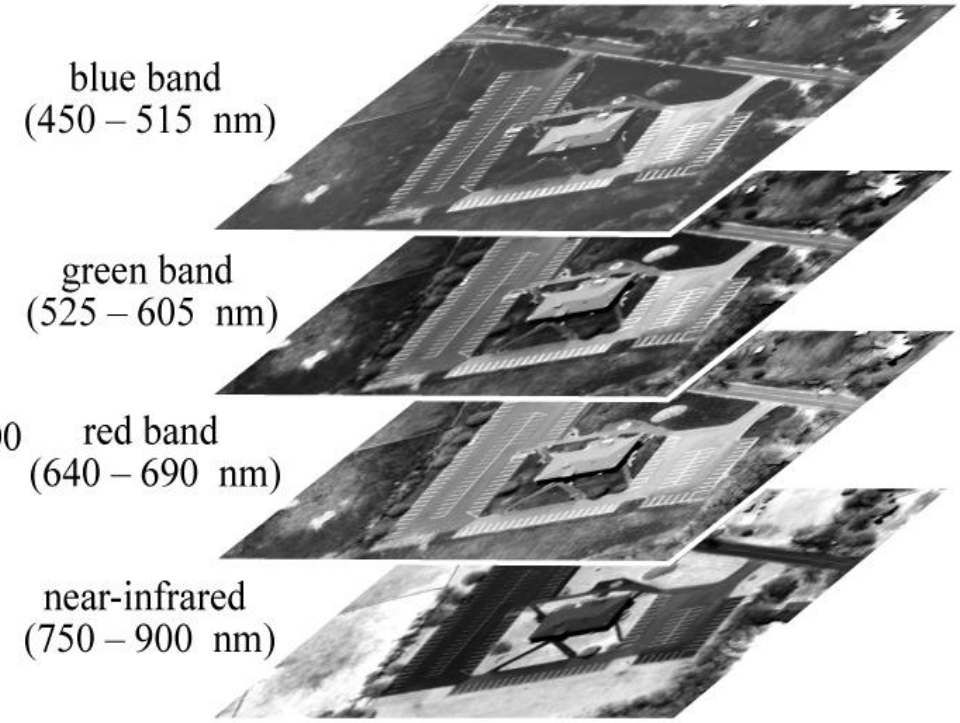


Spectral Resolution

Spectral Resolution of A Typical Multispectral Digital Frame Camera



a. Spectran bands.



b Multispectral remote sensing.

SPATIAL Vs SPECTRAL

- **When extracting urban land cover information from remotely sensed data, analysts tend to consider spatial resolution to be more important than spectral resolution**
- **In other words, it is more useful to have finer spatial resolution than higher spectral resolution**

RADIOMETRIC RESOLUTION

- **The ability of an imaging system to record many levels of brightness, coarse radiometric resolution records very few brightness levels or a few bits, whereas fine resolution records the same scene using many brightness levels**
- **Number of distinguishable gray levels**
- **The finer the radiometric resolution of the sensor, the more sensitive it is to detecting small differences in reflected or emitted energy**
- **The greater the number of levels, the greater the detail of the information**

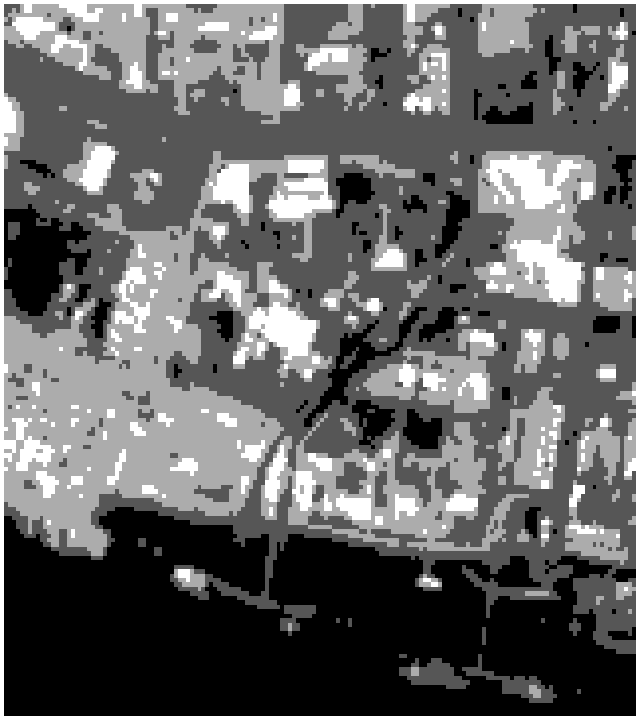
RADIOMETRIC RESOLUTION

- **Number of distinguishable gray levels**
- **Brightness value range**
  - **If 1 bit data, 2 tones: Black (0) and White (1)**
  - **If 2 bit data, 4 tones: Black (0) and White (3)**
  - **If 4 bit data, 16 tones: Black (0) and White (15)**
  - **If 5 bit data, 32 tones: Black (0) and White (31)**
  - **If 6 bit data, 64tones: Black (0) and White (63)**
  - **If 7 bit data, 128 tones: Black (0) and White (127)**
  - **If 8 bit data, 256 tones: Black (0) and White (255)**
  - **If 11 bit data, 2048 tones: Black (0) and White (2047)**

RADIOMETRIC RESOLUTION



1-bit quantization (2 levels)



2-bit quantization (4 levels)

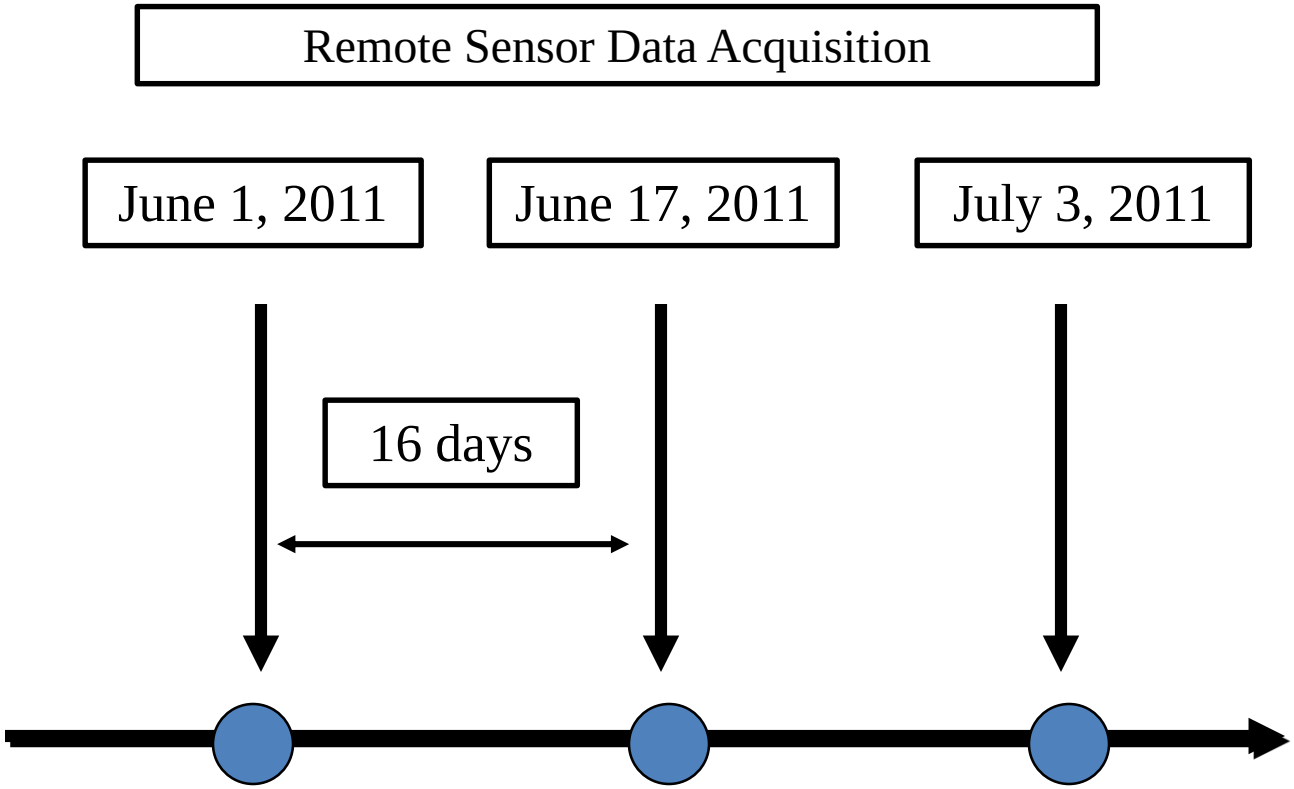


3-bit quantization (8 levels)

TEMPORAL RESOLUTION

- **How often a given place on the earth is revisited (imaged) by a remote sensing system**
- **Temporal resolution of RS satellites is fixed depending upon the orbital parameters**
- **Each satellite has its own unique revisit schedule for obtaining imagery of a particular area**
- **If a satellite imaged the same area every 16 days, then its temporal resolution would be 16 days**

Temporal Resolution



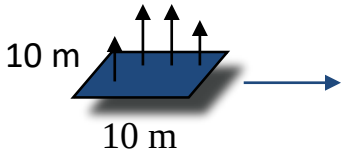
TEMPORAL RESOLUTION

- Temporal resolution of a sensor depends on
  - Satellite/sensor capabilities
  - The swath overlap
  - Latitude
- An important factor to be considered for change detection studies, for short as well as long –lived phenomena
- Aerial imaging , temporal resolution is flexible , rather weather dependent

TEMPORAL RESOLUTION

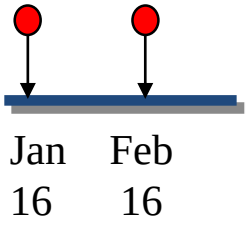
- **With moving sensor, the revisit time can be less as compared to fixed sensors (one to five days)**

Remote Sensor Resolution Considerations





8-bit
(0 - 255)
10-bit
(0 - 1023)



- Spatial - the size of the field-of-view, e.g., 10 × 10 m.
- Spectral - the *number* and *size* of spectral regions the sensor records data in, e.g. blue, green, red, near-infrared, thermal infrared
- Radiometric - sensitivity of detectors to small difference in electromagnetic energy
- Temporal - how often the sensor acquires data, e.g., every 30 days.

?